KWANGWOON UNIVERSITY

Temporal–Channel Input Adaptation for Frozen Time-Series Foundation Models

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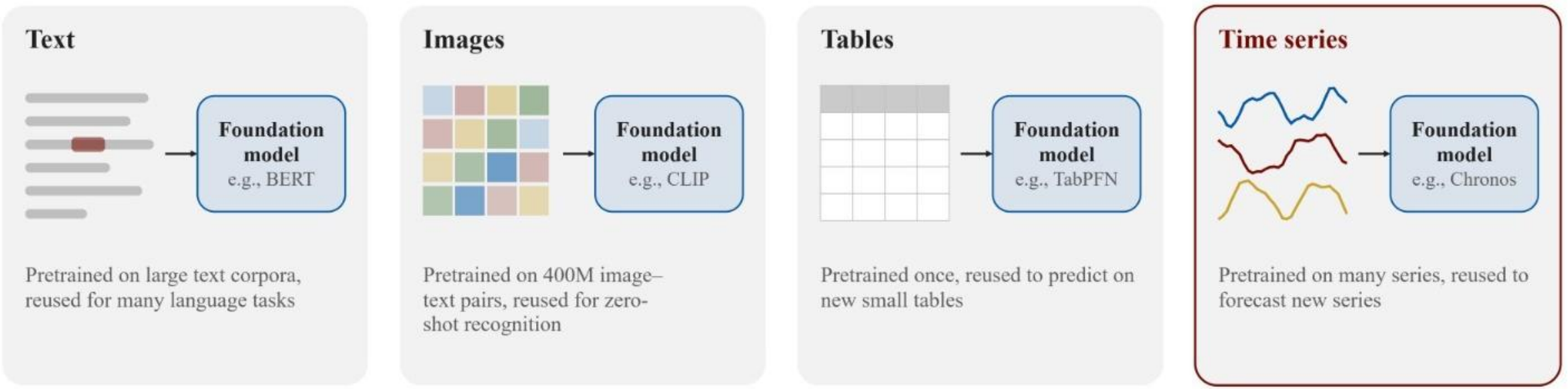
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informs

Background

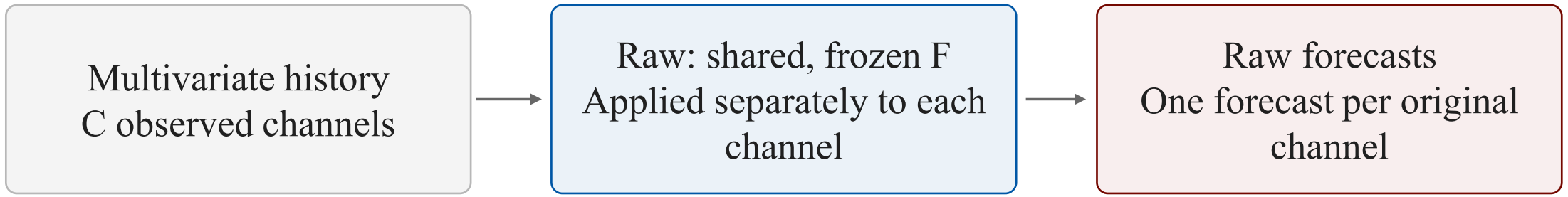
- **Foundation Models (Pretrain Once, Adapt to Many Tasks)**
  - Trained on **broad data at scale**, adaptable to a **wide range of downstream tasks**
  - Available across data types: **BERT** (text), **CLIP** (images), **TabPFN** (tables)
  - **Time-series foundation models (TSFMs)**, e.g., **Chronos**, forecast **new series zero-shot**



R. Bommasani, D. A. Hudson, E. Adeli, et al., On the opportunities and risks of foundation models, arXiv:2108.07258 (2021).
N. Hollmann, S. Müller, L. Purucker, et al., Accurate predictions on small data with a tabular foundation model, Nature 637 (8045) (2025) 319–326.
A. F. Ansari, L. Stella, A. C. Turkmen, et al., Chronos: Learning the language of time series, TMLR (2024).

Background

- **Problem 1: Reusing a Frozen Univariate Forecaster on Multivariate Data**
  - Setting: reuse the pretrained forecaster without updating its weights.
  - Raw applies the same forecaster to each channel separately; it has no cross-channel input.
  - Goal: adapt to the target data while balancing forecast error and inference cost.



Cross-channel dependence is a motivation to test—not a guarantee of additional predictive information.

Ansari et al., Chronos, TMLR (2024). Benechehab et al., AdaPTS, ICML (2025).

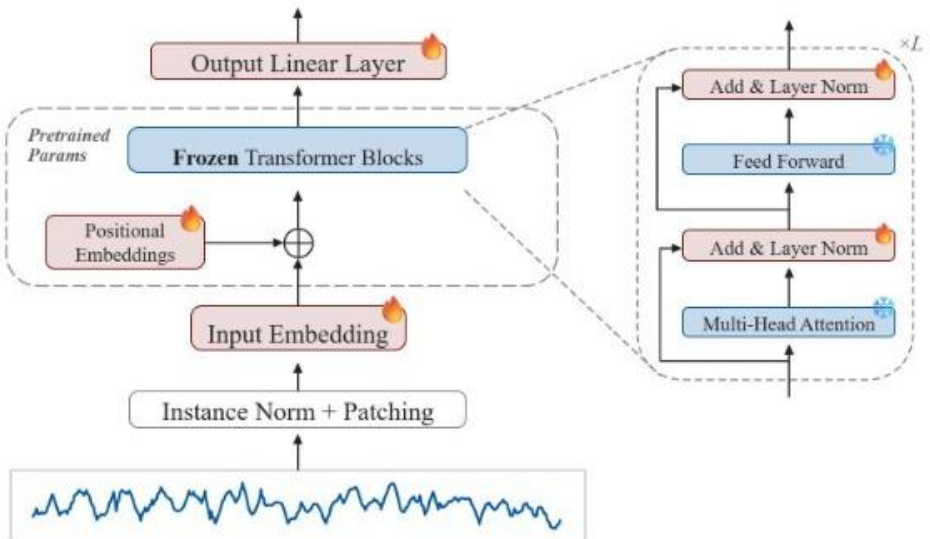
Background

- **Adapting the TSFM Itself (Tuning Inside the Model)**

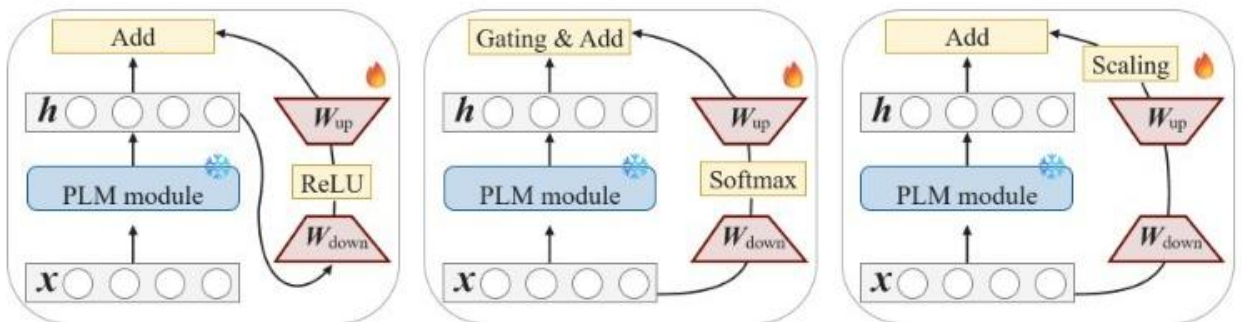
trainable frozen

  - Full fine-tuning retrains every weight: **a new model for every task**
  - Partial fine-tuning and PEFT train **only a small part**: selected layers or added modules
  - Both work inside the model, changing **hidden states of specific layers**

Partial fine-tuning



PEFT: adapter, prefix tuning, LoRA



(a) Adapter (b) Prefix Tuning (c) LoRA

N. Houlsby, A. Giurgiu, S. Jastrzebski, et al., Parameter-efficient transfer learning for NLP, in: Proc. ICML, 2019, pp. 2790–2799.
T. Zhou, P. Niu, X. Wang, et al., One fits all: Power general time series analysis by pretrained LM, in: Proc. NeurIPS, 2023.
J. He, C. Zhou, X. Ma, et al., Towards a unified view of parameter-efficient transfer learning, in: Proc. ICLR, 2022.

Background

- **External Adaptation: Learn the Interface, Keep the Forecaster**
  - An input adapter mixes the original channels into K latent series.
  - The same frozen F predicts each latent series; an output adapter returns C channels.
  - K < C reduces the number of series sent through F; total speed is measured end to end.

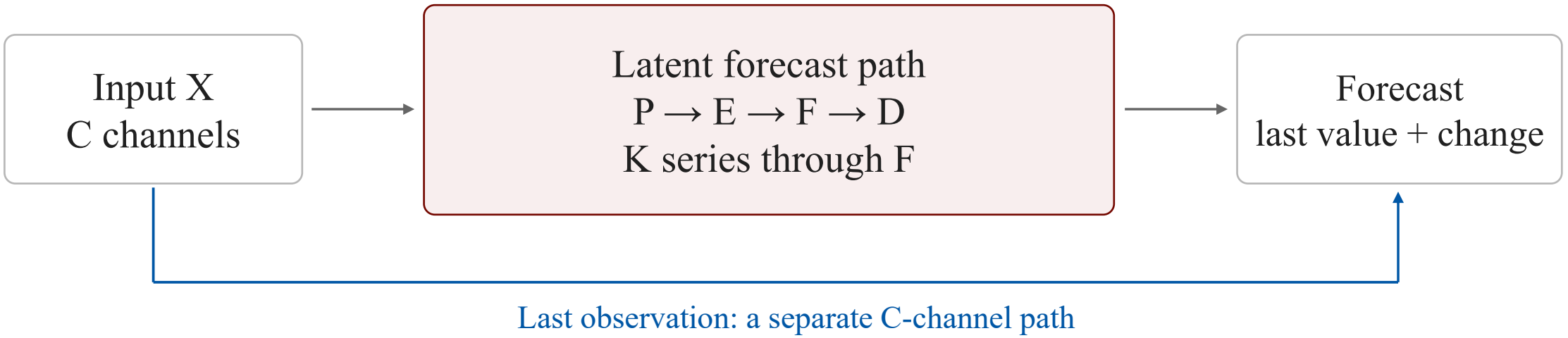


Frozen weights preserve the checkpoint; they do not make adapter training free.

Benechehab et al., AdaPTS: Adapting Univariate Foundation Models, ICML (2025).

Problem

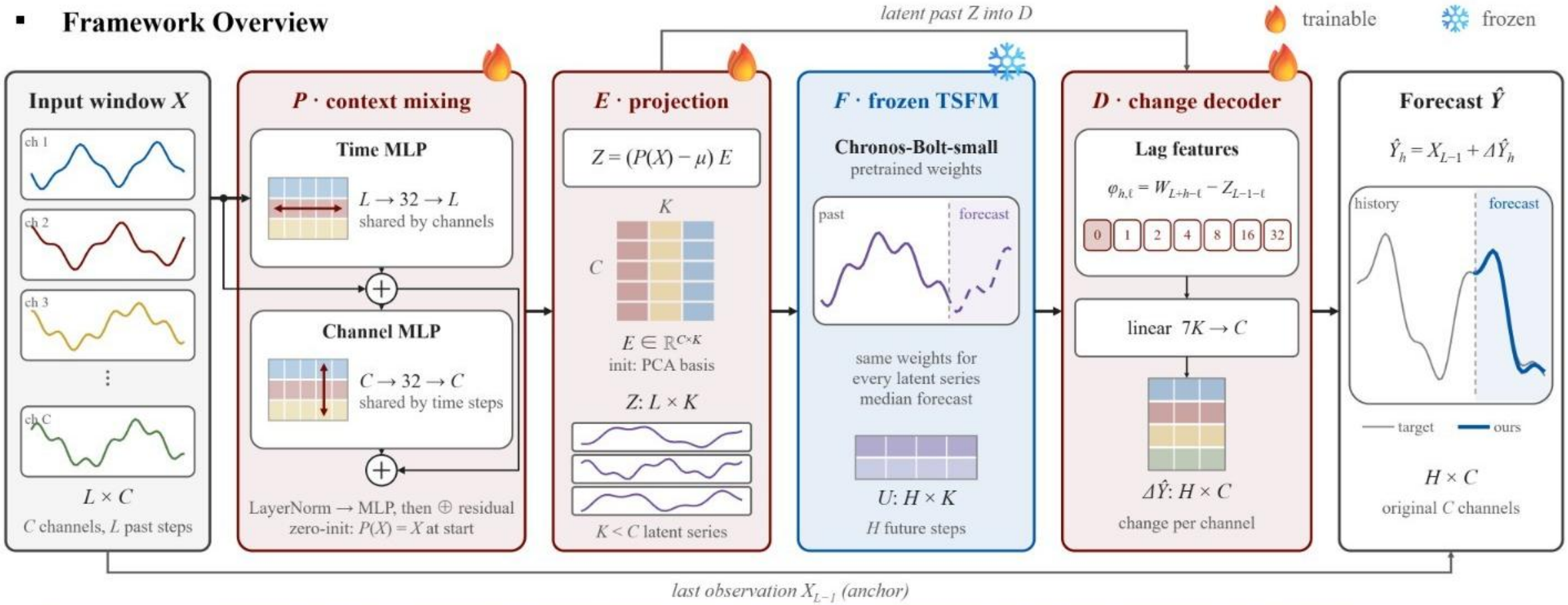
- **Problem 2: Limited Latent Capacity Must Carry Useful Forecast Changes**
  - A small K limits the representation available to the latent forecasting path.
  - Input reconstruction variance is not the same as information needed to predict the future.
  - Our design preserves the last observed level separately and learns forecast changes.



Eckart and Young, Psychometrika (1936): reconstruction result only. Benechehab et al., ICML (2025).

Proposed Method

- **Framework Overview**



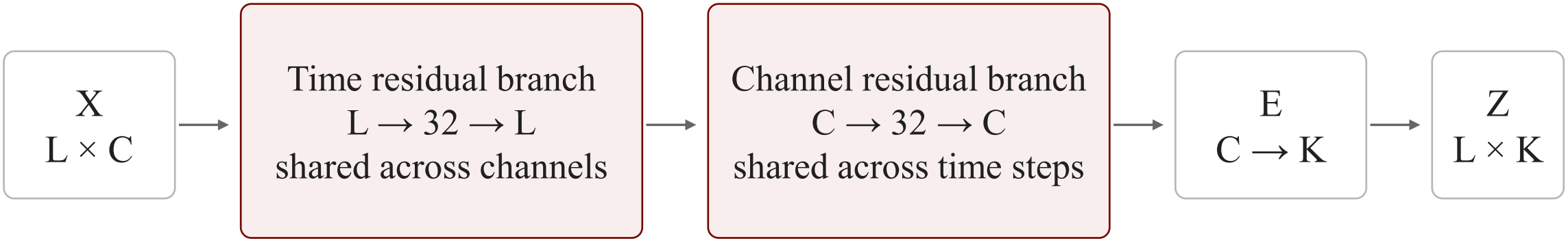
Goal: reuse frozen F with fewer latent series and a useful accuracy–cost trade-off

P adjusts context before projection; the last observed level bypasses the latent forecast path.

S.-A. Chen, C.-L. Li, N. Yoder, et al., TSMixer: An all-MLP architecture for time series forecasting, TMLR (2023).
A. Benechehab, V. Feofanov, G. Paolo, et al., AdaPTS: Adapting univariate foundation models to probabilistic multivariate time series forecasting, in: Proc. ICML, 2025, pp. 3731–3748.
A. F. Ansari, L. Stella, A. C. Turkmen, et al., Chronos: Learning the language of time series, TMLR (2024).

Proposed Method

- **Context Module P: Adjust the Representation Before Compression**
  - F reads an already compressed history; P can use the full observed window before E.
  - E is a shared linear projection; P adds nonlinear, input-dependent context processing.
  - This motivates the design; it does not prove that both branches are necessary.



Residual outputs start at zero: P(X) = X initially. Joint LayerNorm still couples time and channels.

Chen et al., TSMixer, TMLR (2023). Current implementation: temporal + channel residual MLPs.

Proposed Method

- **Projection E: A Reproducible Rule for Initial Capacity K**
  - Choose the smallest K retaining at least $\tau = 0.9$ of the A_fit input PCA variance.
  - Initialize E from PCA, then update E with the final forecast loss.
  - This is an initial-capacity heuristic—not a guarantee of optimal predictive dimension.

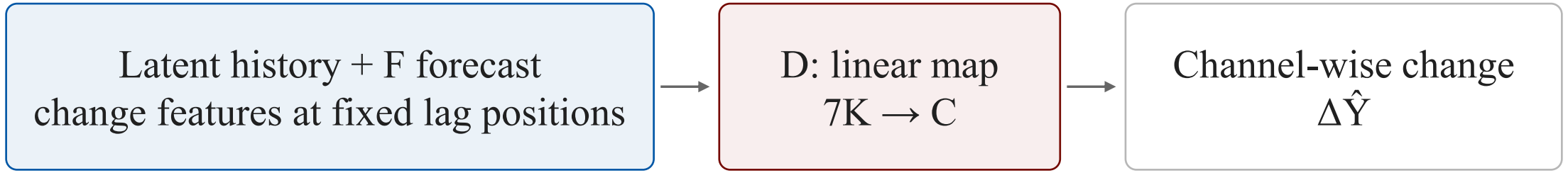
Dataset	C	K at $\tau = 0.9$	**A_fit PCA variance retained**
Jena	21	6	91.1%
Appliances	26	10	90.7%
ETTh1	7	4	93.0%
ETTh2	7	4	93.6%

K uses A_fit (0–50%) PCA variance. Refit PCA/normalization use A60 (0–60%); K is unchanged in all four datasets.

Source: k_tau_rule_20261023_v1/K_TAU.csv, $\tau = 0.9$, cumulative_A_fit. Rea and Rea (2016): heuristic context.

Proposed Method

- **Decoder D: Preserve the Current Level and Predict a Change**
  - The last observed value reaches the output directly in all C original channels.
  - D combines latent change features at several past reference positions.
  - Lag positions {0, 1, 2, 4, 8, 16, 32} are a fixed design choice; their added value is not yet isolated.



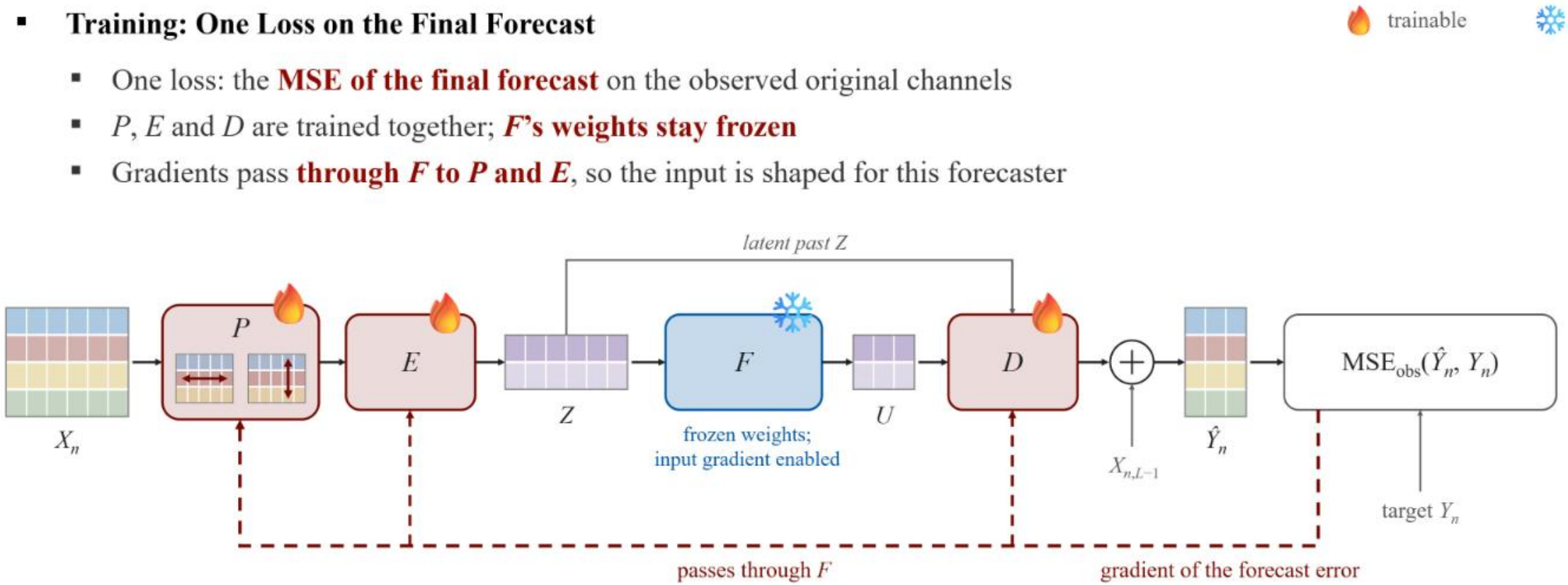
Forecast = last observed value + predicted change

Illustration: 15.0 °C + 0.8 °C = 15.8 °C

Zeng et al., LTSF-Linear / NLinear, AAAI (2023): last-value residual formulation. Registered D implementation.

Proposed Method

- **Training: One Loss on the Final Forecast**
  - One loss: the **MSE of the final forecast** on the observed original channels
  - *P*, *E* and *D* are trained together; ***F*'s weights stay frozen**
  - Gradients pass **through *F* to *P* and *E***, so the input is shaped for this forecaster



A. Benechehab, V. Feofanov, G. Paolo, et al., AdaPTS: Adapting univariate foundation models to probabilistic multivariate time series forecasting, in: Proc. ICML, 2025, pp. 3731–3748.
M. Jin, S. Wang, L. Ma, et al., Time-LLM: Time series forecasting by reprogramming large language models, in: Proc. ICLR, 2024.

Experiments

- **Experimental Setup**
  - **Three experiments**
    - **Comparative study**: Raw, AdaPTS adapters and ours
    - **Ablation**: change one module (P or D) at a time
    - **Sensitivity**: the τ rule's K versus other widths
  - **Forecaster**
    - Chronos-Bolt-small, weights frozen
  - **Metric**
    - V MSE: per-channel MSE on split V, averaged over channels, on data scaled with training statistics
    - Also MAE, MASE, R^2 and MAPE, all on V
  - **Runs**
    - 10 seeds per trained method; same split, K and epoch selection
    - V: development split, not an independent test

Dataset	Domain	$C \rightarrow K$	$L \rightarrow H$	V origins
Jena	Weather	21 → 6	512 → 48	43
Appliances	Energy	26 → 10	512 → 48	41
ETTh1	Electricity	7 → 4	512 → 48	36
ETTh2	Electricity	7 → 4	512 → 48	36

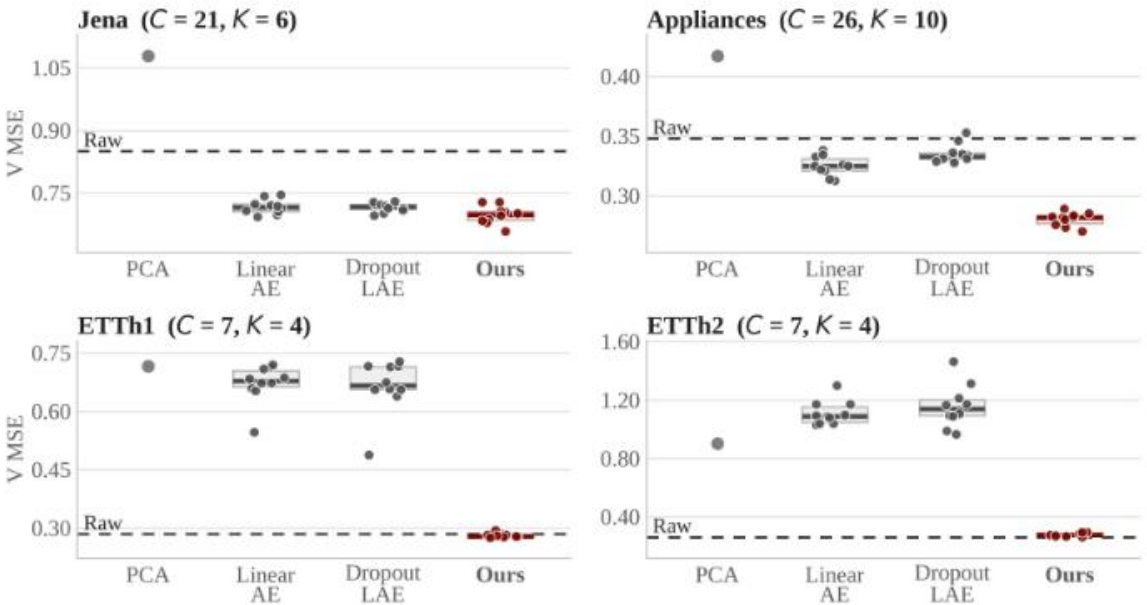
K: set before training by the $\tau = 0.9$ rule (slide 9). ETTh2: first 17,400 hours. Time-ordered splits: training 0–60% (epochs chosen on 50–60%), V 70–80%.

- **Why Chronos-Bolt-small?**
  - Univariate, as in Problem 1: one series at a time
  - Real values in, quantiles out: gradients pass through F
  - Small (48M parameters); its model card reports it faster and more accurate than the original Chronos

A. F. Ansari, L. Stella, A. C. Turkmen, et al., Chronos: Learning the language of time series, TMLR (2024).
Amazon, chronos-bolt-small model card, Hugging Face, https://huggingface.co/amazon/chronos-bolt-small.

Experiments

- **Comparative Study: Raw, AdaPTS Adapters and Ours**
  - All methods: the same frozen F, data, K and epoch selection; **10 seeds** per trained method
  - Ours vs Raw (median MSE): Jena **−17.8%**, Appliances **−18.9%**, ETTh1 **−1.3%**, ETTh2 +6.8%
  - Trained AdaPTS adapters beat Raw only on Jena and Appliances; **ours is the best adapter on all four** (a tie on Jena)



Median V MSE over 10 seeds (lower is better)

AdaPTS adapters (ICML 2025)

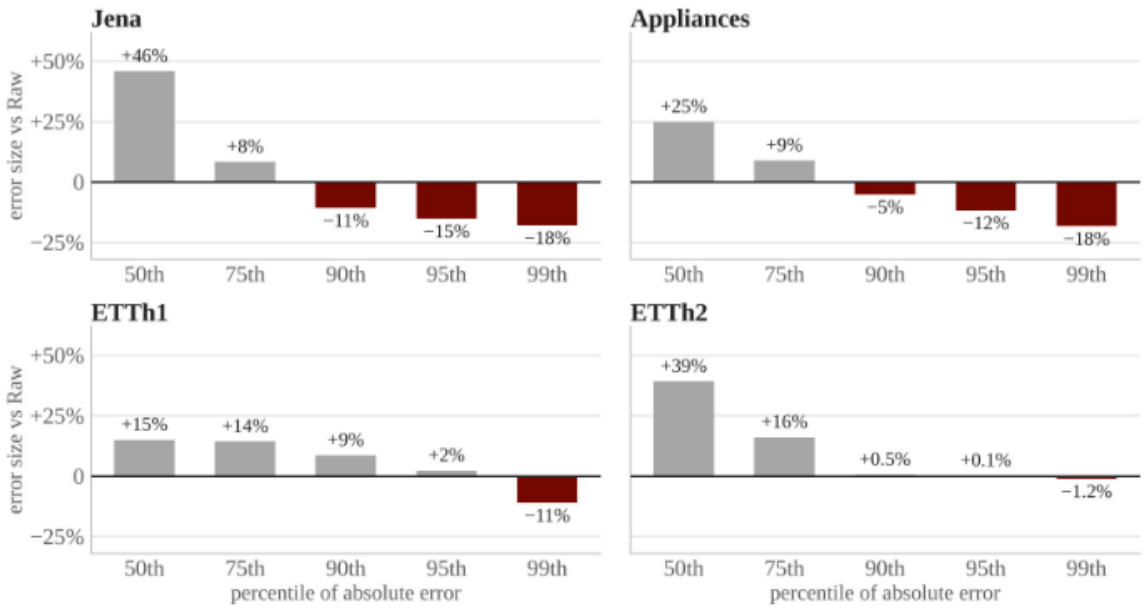
Data	Raw	PCA	LinearAE	dropoutLAE	Ours
Jena	0.849	1.079	0.715	0.716	0.698
Appliances	0.348	0.417	0.325	0.333	0.282
ETTh1	0.283	0.716	0.678	0.666	0.280
ETTh2	0.260	0.902	1.091	1.137	0.277

AdaPTS adapters: ported to our setup, not the paper's; results are provisional.

A. Benechehab, V. Feofanov, G. Paolo, et al., AdaPTS: Adapting univariate foundation models to probabilistic multivariate time series forecasting, in: Proc. ICML, 2025, pp. 3731–3748.

Experiments

- **Comparative Study: Error Distribution and Other Metrics**
  - MSE squares each error, so **large errors count more** than they do in MAE
  - Ours vs Raw: **MSE lower** on three datasets, but **MAE higher** on all four
  - Error quantiles: q90/q95/q99 lower on Jena/Appliances; on ETT, only q99 is lower among those shown.



Ours vs Raw, median of 10 seeds

Metric	Jena	Appliances	ETTh1	ETTh2
MSE	−17.8%	−18.9%	−1.3%	+6.8%
MAE	+3.3%	+1.2%	+7.5%	+14.4%
MASE	+10.7%	+0.1%	+11.5%	+23.5%
R^2	0.56 / 0.50	0.52 / 0.47	0.37 / 0.44	0.26 / 0.42
MAPE	−5.3%	−5.7%	+7.8%	+14.8%

R^2: ours / Raw; other rows: % change.
MAPE in original units: $|y| \geq 0.1 \times \sigma_A60,c$.

R. J. Hyndman, A. B. Koehler, Another look at measures of forecast accuracy, International Journal of Forecasting 22 (4) (2006) 679–688.

Experiments

- **Cost: Fewer Series Through the Frozen TSFM**
  - *F* forecasts ***K* latent series instead of *C* channels**; only the small adapter is trained
  - Jena, same rows per *F* call: about **3.6× faster** than Raw and LoRA
  - On accuracy alone, LoRA and Ridge are better on Jena, and Direct MLP on Appliances and ETTh2

Cost per dataset (memory and speed-up: ours vs Raw)

Data	C → K	Trainable parameters, ours	Trainable parameters, AdaPTS-LinearAE	Memory vs Raw	Speed-up vs Raw
Jena	21 → 6	35,717	279	−26.4%	2.07×
Appliances	26 → 10	37,114	556	not measured	not measured
ETTh1	7 → 4	34,023	67	−10.1%	1.30×
ETTh2	7 → 4	34,023	67	−10.1%	1.26×

Inference cost: batch 8; median latency over 20 repeats; OURS = mean of two seeds.
Table: default ≤128 F rows/call. Jena 3.6×: separate fixed-48-row setting. Memory: peak allocated GPU memory.

Experiments

- **Ablation: Component Roles and the Jena P-Branch Follow-up**

Registered ablations: change in median V MSE

10 seeds; positive = worse after removal/replacement

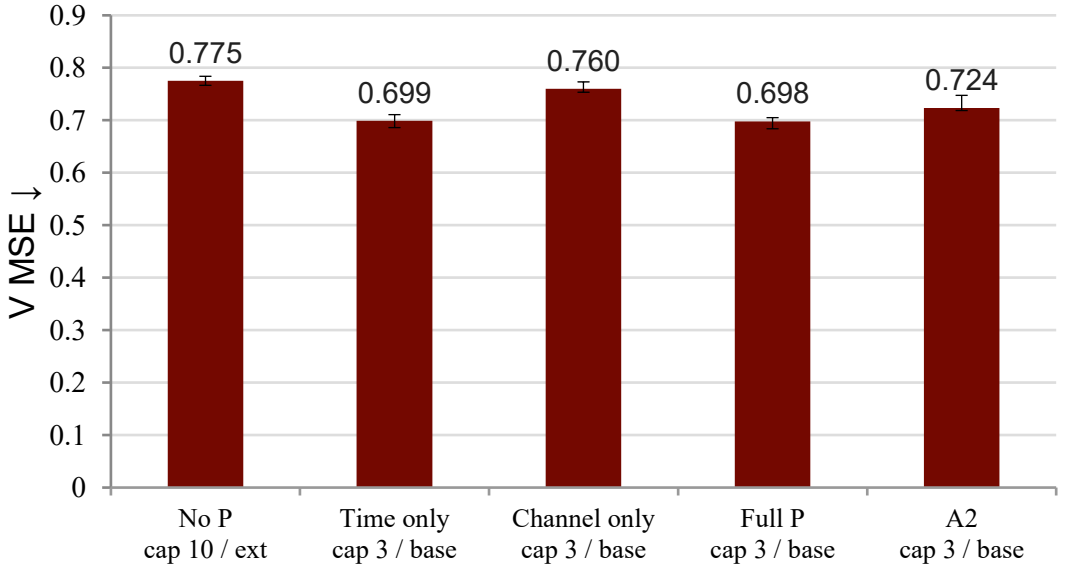
Dataset	No P	Linear D
Jena	+11.12%	−1.75%
Appliances	+0.56%	+18.57%
ETTh1	−0.15%	+155.95%
ETTh2*	−5.00%*	+225.34%

P contributes on Jena; D contributes on Appliances and ETTh1 under the registered rule.

* ETTh2 is auxiliary; its No-P result remains provisional.
A2 (parameter-count reference): UNCLEAR on all four datasets.

Jena: exploratory P-branch comparison

10 seeds; median V MSE, whiskers Q1–Q3; lower is better



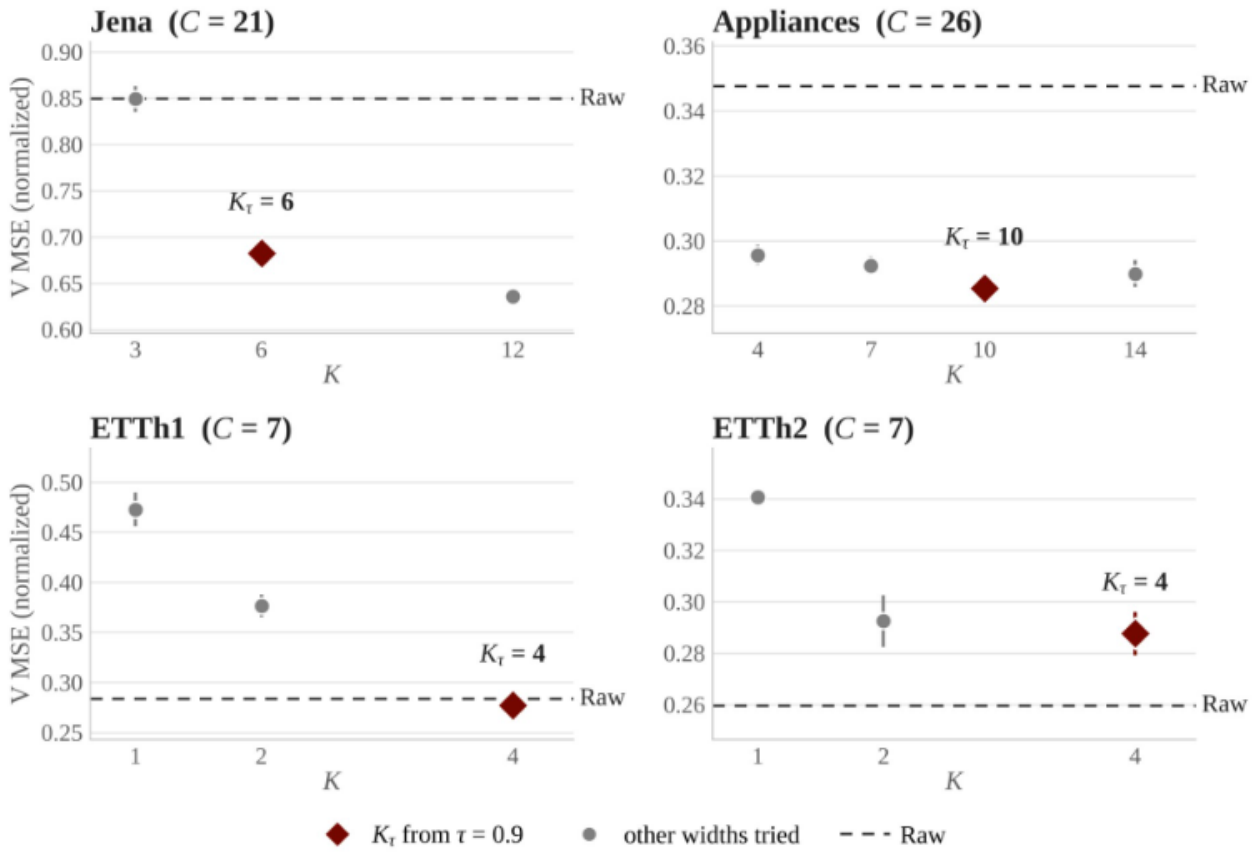
Time only +0.17%; channel only +8.92% vs full P.

Unequal branch capacity: time MLP 33,312; channel MLP 1,397. Branch effects ≠ intrinsic superiority of an axis.

Sources: comparative_ablation_followup_20261027_v1; jena_p_branch_20261027_v1. V = development validation.

Experiments

- **Sensitivity to K: The τ Rule vs. Other Widths**
  - **Setup**
    - K: set by the $\tau = 0.9$ rule (slide 9) before training
    - Other widths: earlier runs, same recipe
    - Two-seed means; bars: min–max
  - **Result**
    - Appliances, ETTh1: no other width tried is better
    - Jena: **$K = 12$ beats the rule's $K = 6$** by more than the seed spread
    - ETTh2: every width is worse than Raw
    - Measured under default batching:
      217 / 243 / 294 MiB at K = 3 / 6 / 12.
      Fixed 48 rows/call: K = 12 ≈ 243 MiB.



Conclusion

- **What the Evidence Supports—and What Remains Open**

Design	Observed benefit	Component evidence
Adapt before compression; preserve the current level separately	Raw MSE reduced on Jena (17.8%) and Appliances (18.9%)	P contributes on Jena; D contributes on ETTh1 and Appliances

Jena: vs time-only, unclear; vs channel-only, full P lower.

- Scope: ETTh2 MSE and all four MAE results remain worse than Raw; comparator wins are provisional.
- K from PCA is a heuristic: Jena benefits from a larger tested K. Accuracy–cost trade-offs are conditional.
- Next: isolate the value of F, finish comparator optimization, then evaluate a frozen design independently.

All accuracy claims here are development-V results. Comparator extensions and independent test remain open.

Thank you!